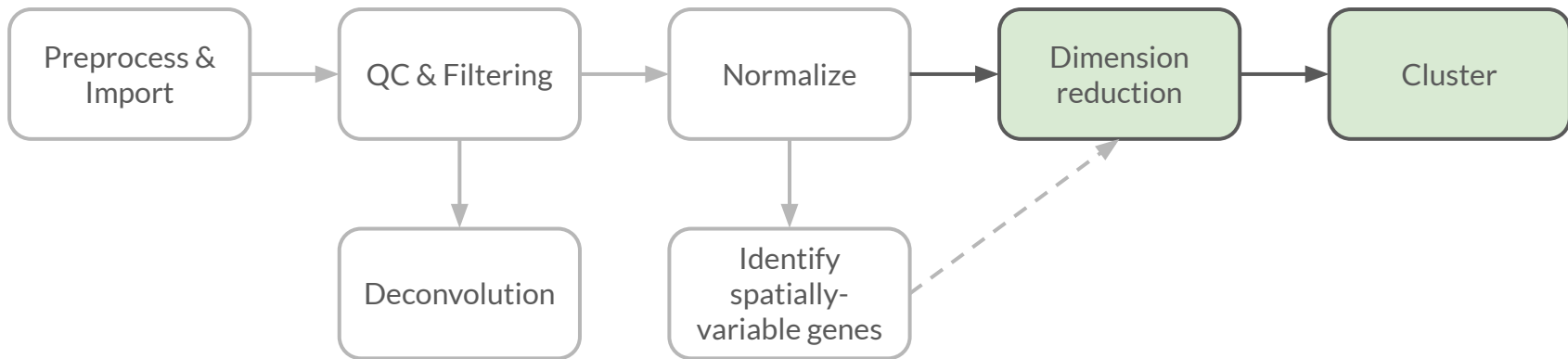
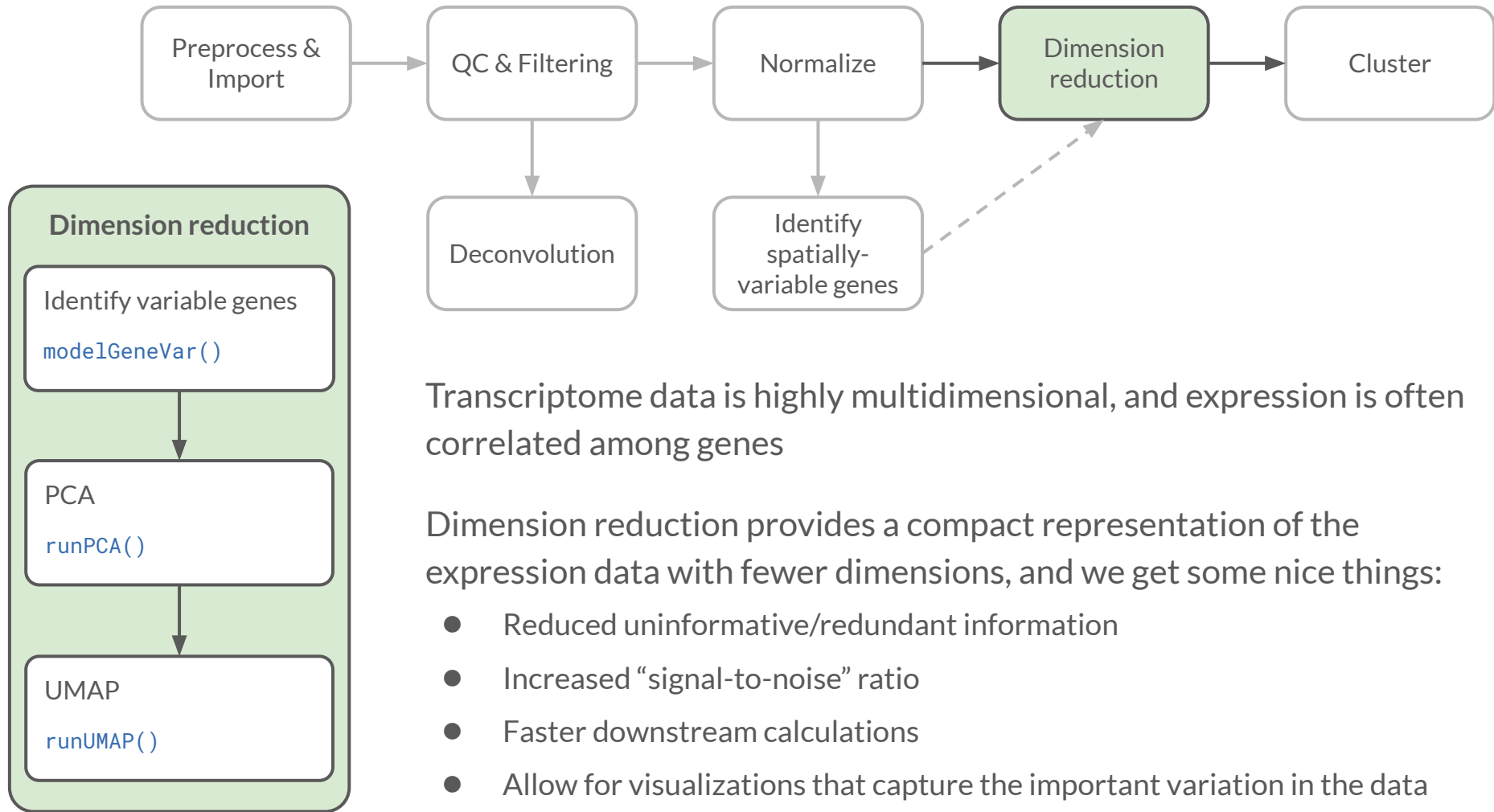


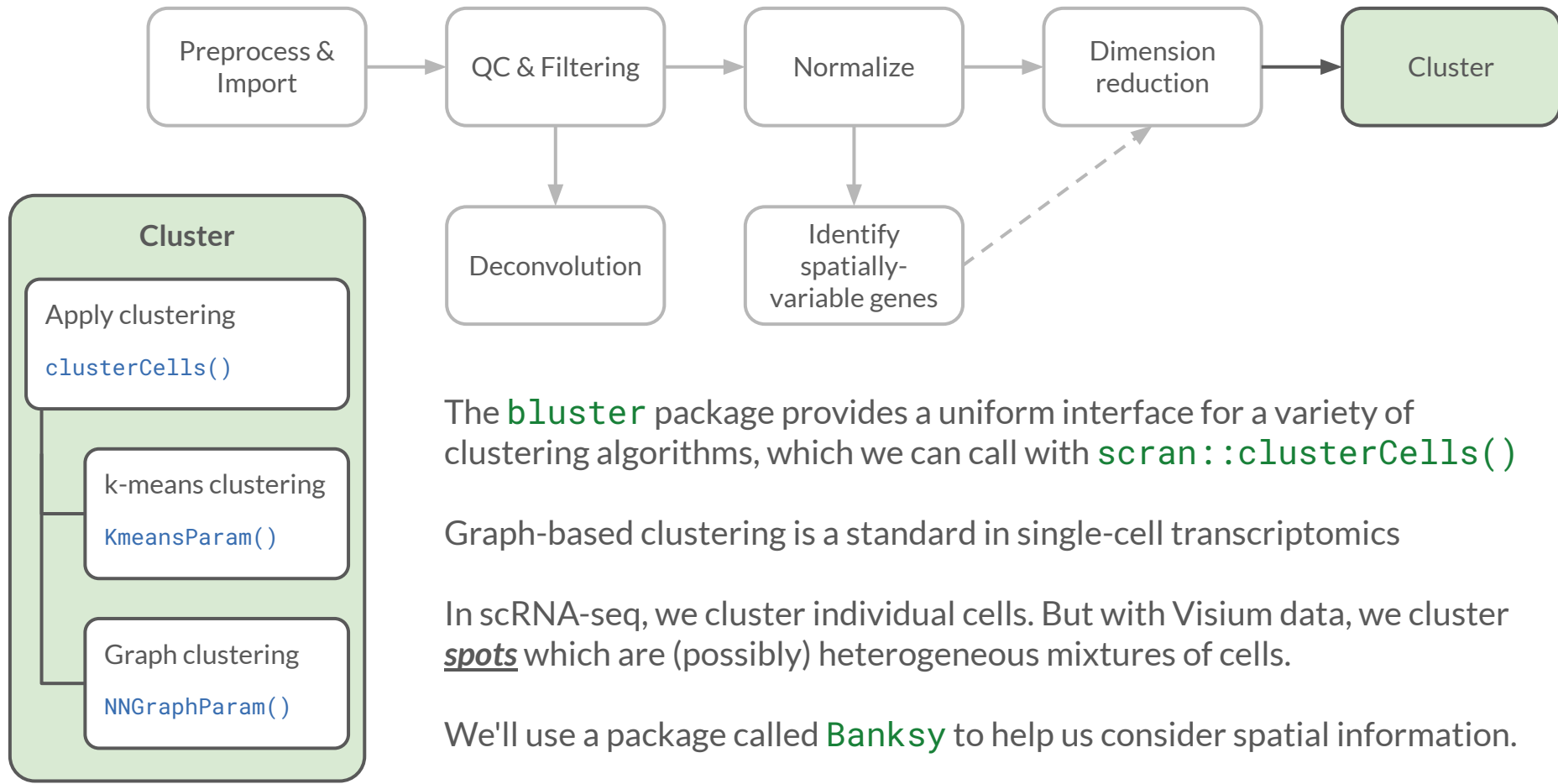


Clustering spatial transcriptomics data

Data Lab







What's a domain, anyways?

You'll often see these terms in the literature. Here's how we'll use them:

Cluster: Mathematical grouping of (here) spots that we hope is biologically meaningful

Domain: A spatially defined region consisting of relatively consistent gene expression profiles (source: [OSTA](#))

Region: A more generic term for an area of interest in the tissue

Clustering is one way that we can identify spatial domains, ideally using multiple sources of information, e.g. **logcounts** and spatial information such as distances between spots

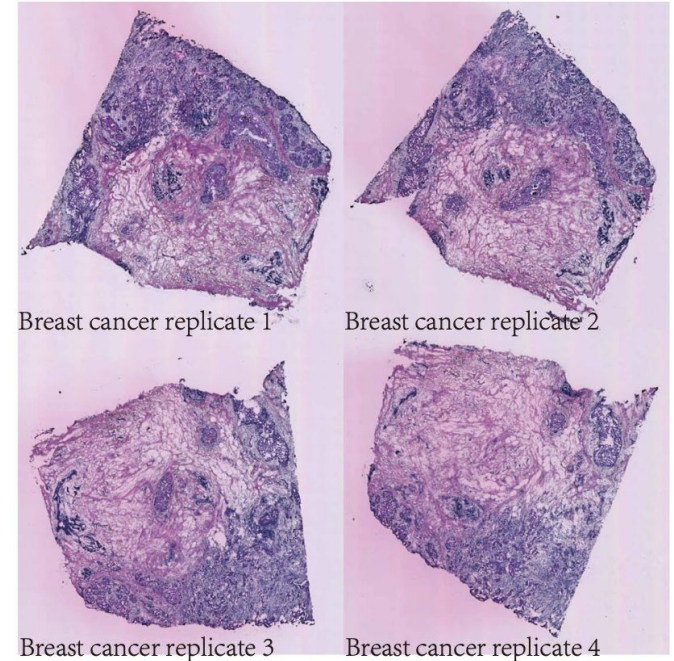


Caution: Distance and coordinates are different!

The tissue orientation (top, bottom, left, right...) is *generally meaningless*

Calculations should not use absolute locations to represent spatial information, but instead a *rotation-invariant metric* (e.g. distance)

In other words, the calculations should not change if the tissue is rotated.



<https://doi.org/10.1038/s41467-025-62574-4>

We'll be using a method called Banksy

Building Aggregates with a Neighborhood Kernel and Spatial Yardstick

(h/t for some solid backronym work)

Runs on different spatial resolutions:

- Visium 55 um: Detect clusters with coherent expression and spatial arrangement
- Visium HD, Xenium, etc: Detect clusters more likely to match individual cell types
 - PSA: We do not recommend assuming 1 cluster = 1 cell type. Perform annotation separately!!

Allows you to weight the relative levels of expression vs. spatial contribution

Highly scalable – does not use a neural network (in other words: no deep learning)



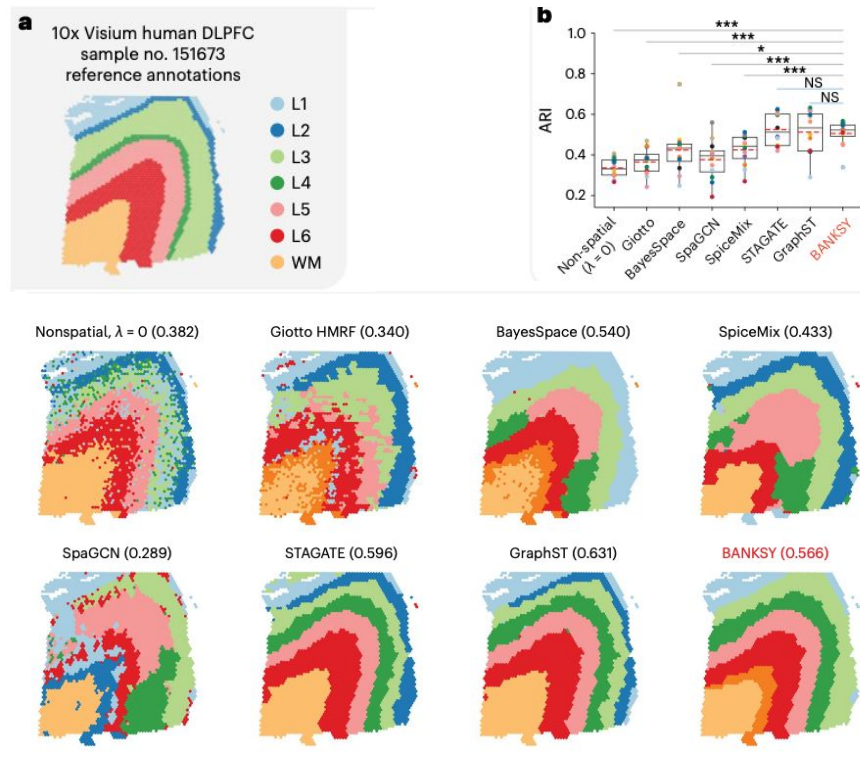
Comparing clustering in a Human DLPFC (dorsolateral prefrontal cortex) Visium dataset

Adjusted Rand Index (ARI) measures the similarity of two data clusterings:

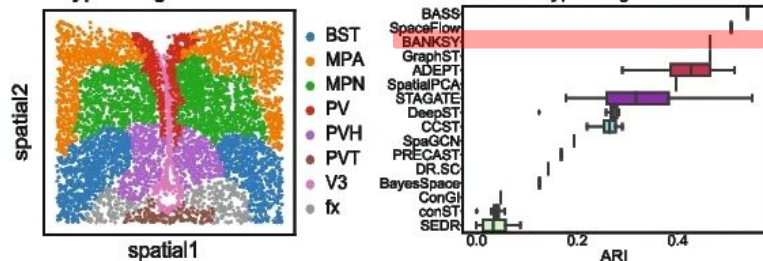
Are cells (spots) grouped together in one set of clusters also grouped together in some other set of clusters?

-1 → Grouped entirely differently

1 → Grouped entirely identically



MHypo Bregma -0.09 GT

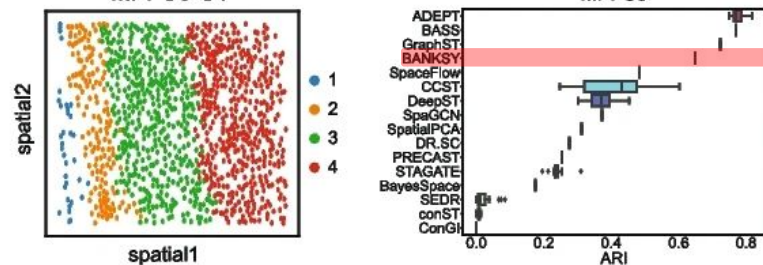


MERFISH
Mouse hypothalamus

Performance on image-based ST

e

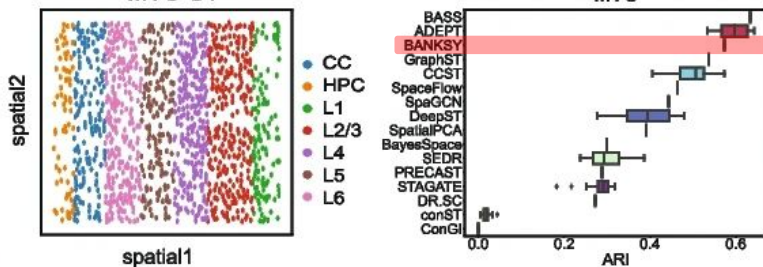
MPFC3 GT



StarMAP
Mouse prefrontal cortex

f

MVC GT



StarMAP
Mouse visual cortex

*Benchmarking clustering, alignment, and
integration methods for spatial
transcriptomics*

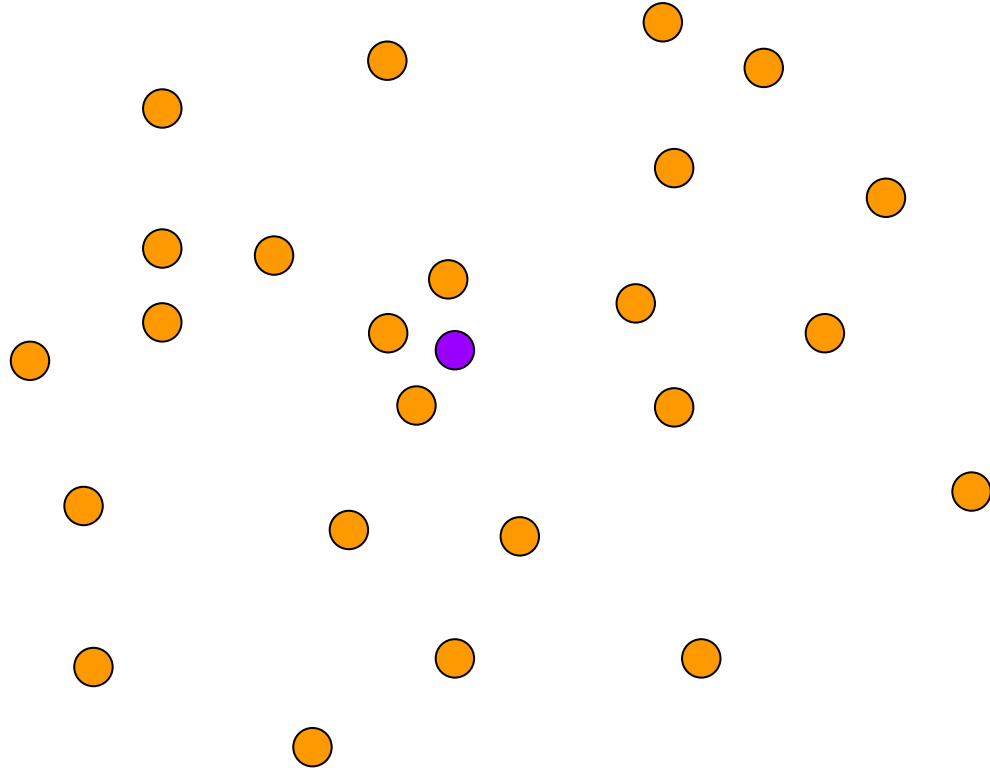
<https://doi.org/10.1186/s13059-024-03361-0>

The Banksy approach

1. Calculate spatially-informed Banksy matrices, starting with a kNN graph
 - a. Weighted mean of gene expression in each cell's neighborhood
 - b. (Optionally) Azimuthal Gabor filter (AGF) adds a directional/gradient component
2. Combine all of these matrices into a single matrix and calculate PCA from the combined matrix
 - a. <Expression + weighted mean matrix> ...or... <Expression + weighted mean matrix + AGF matrix>
3. Cluster that PCA with your favorite algorithm (e.g. Louvain) → Spatially-informed clusters, which might correspond to "spatial domains"

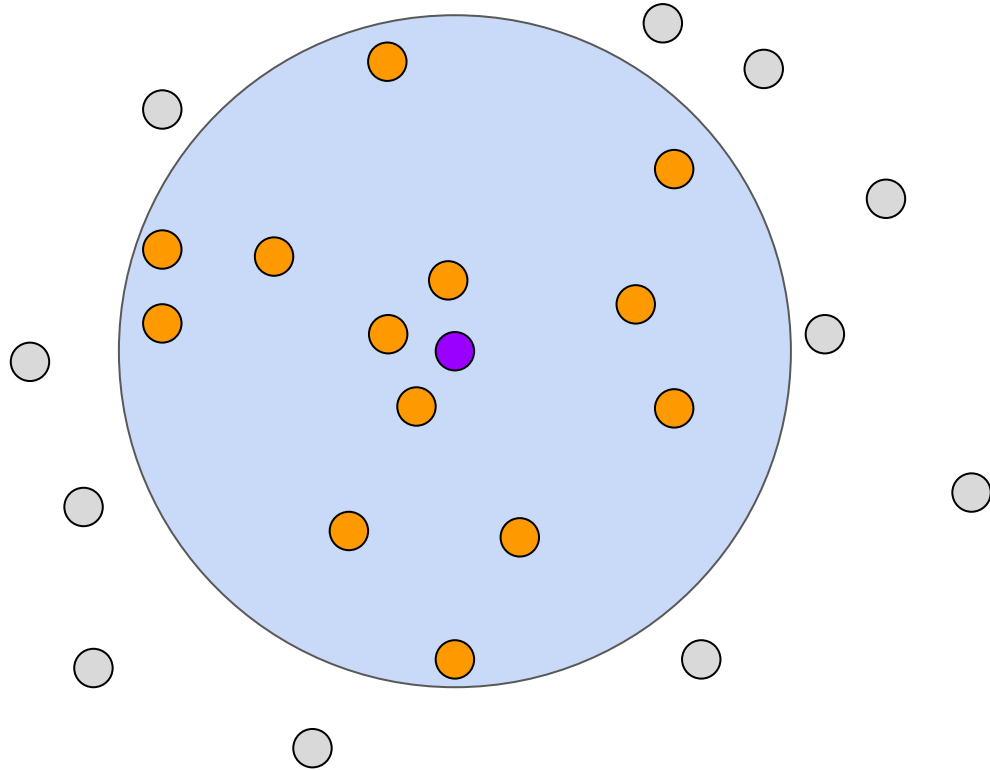
kNN graphs with a spatial kernel

A bunch of
spots around a
target spot



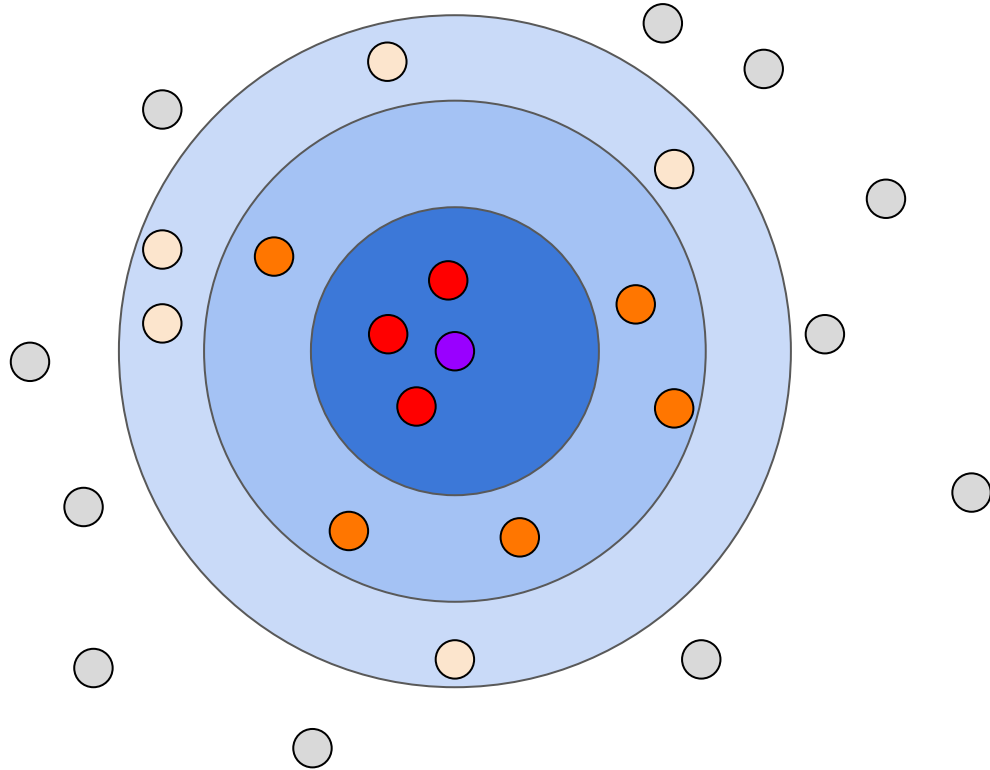
kNN graphs with a spatial kernel

Define a $k=12$
neighborhood



kNN graphs with a spatial kernel

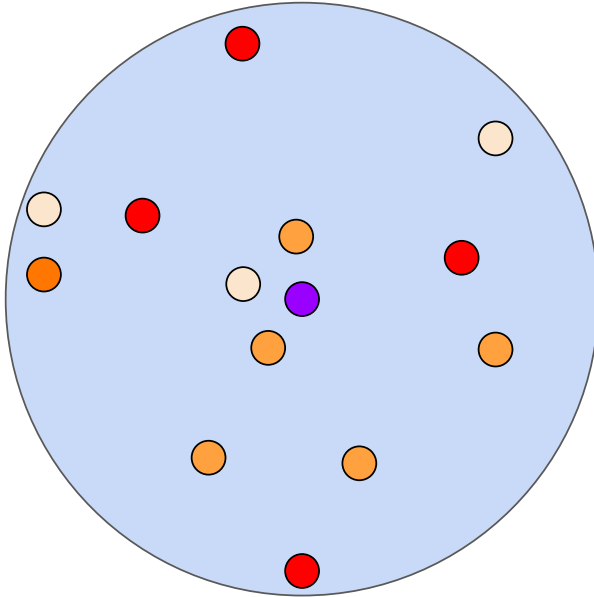
Apply a spatial kernel where spots closer to target get a higher weight



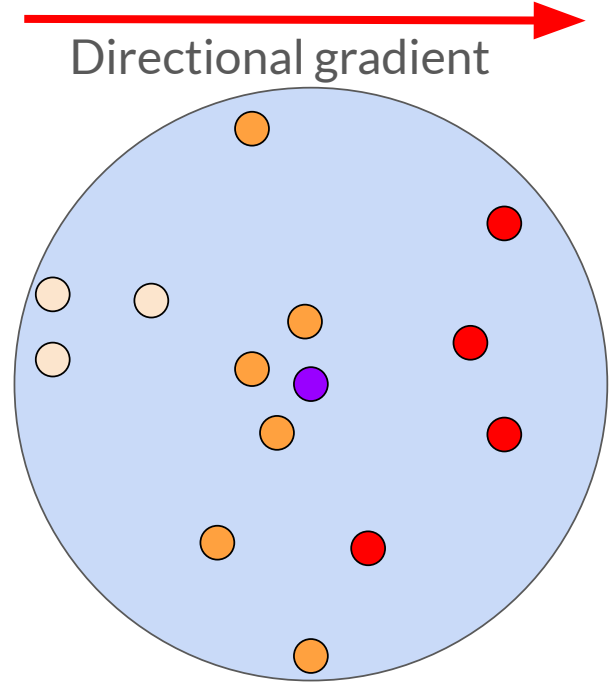
Conceptually, AGF helps Banksy distinguish between these neighborhoods

- High expression
- Medium
- Low

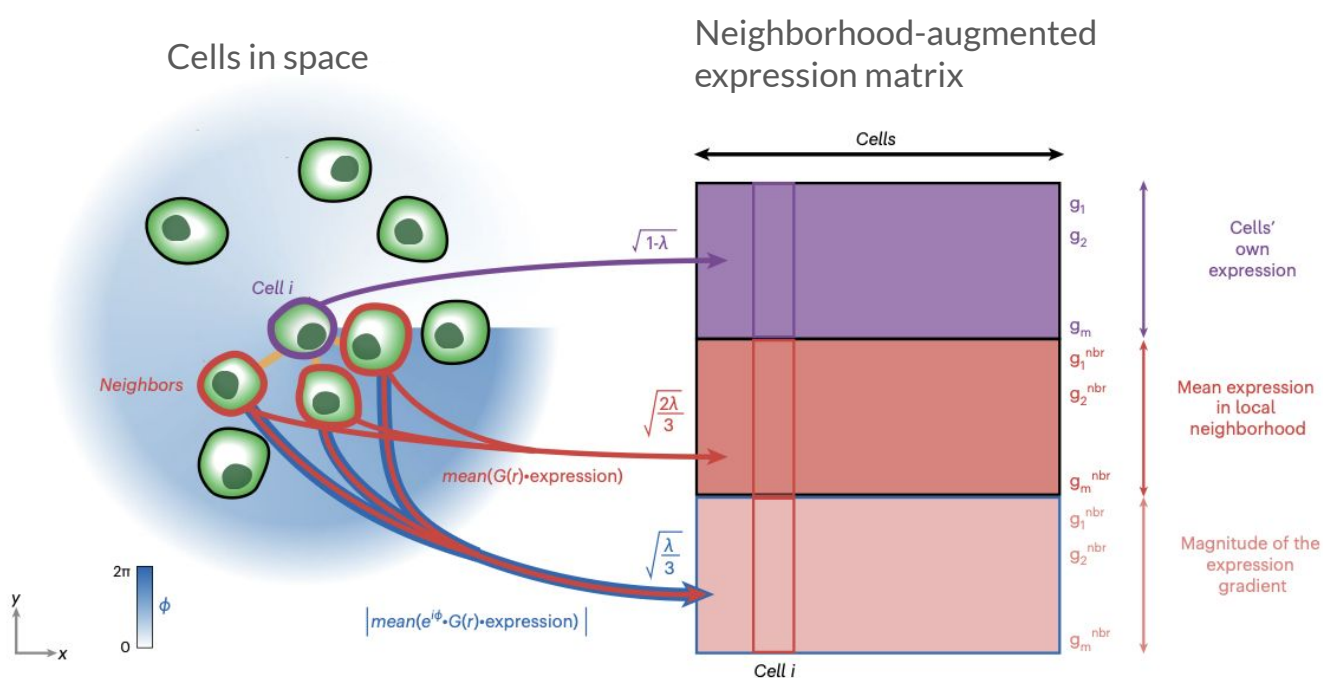
No directional gradient



Directional gradient



Matrices are then *combined* using a tuning parameter λ



$\lambda = 0 \rightarrow$ Expression only
 $\lambda = 1 \rightarrow$ Spatial matrices only

(Note: Image says "cells," but it's just as well for "spots")

